

THE 6 CORE ARCHITECTURE MODULES:

Module 1 SATARK-DRISHTI - - Master Risk Scorer Delay Predictor

CORE ENGINE: Composite Scorer 0-100 Survival Analysis

WHAT IT DOES: Combines findings from all 7 AI sentinels into one single Priority Risk Score between 0 and 100. It also predicts how many months a project will be delayed.

HOW TO EXPLAIN: Think of this like a CIBIL Score for credit cards or a Student Report Card. Instead of handing an officer 7 different complicated reports, it gives one final score e.g. 85/100 - CRITICAL RISK so

EXAMPLE: Combines Statutory Violations +30, CPWD Overpricing +25, and Cartel Risk +25 to assign an 80/100 Critical Risk rating.

Module 2 SATARK-SAMVAAD - - GenAI Natural Language Audit Copilot

CORE ENGINE: GenAI Natural Language Copilot over 176,925 MoSPI Records

WHAT IT DOES: Lets government officers ask plain-language questions in English or Hindi over 176,925 government records instead of writing complex database code.

HOW TO EXPLAIN: Like ChatGPT for government audits An officer can simply type: Show high-risk projects in Bihar under 5 Lakh, and SAMVAAD instantly scans all records and presents the exact matching projects

EXAMPLE: Query: Find ghost assets in Bihar - Returns 42 matching projects with total financial exposure of 1.8 Cr.

Module 3 PRASHNA-KAVACH - - The Why Was This Flagged Evidence Drawer

CORE ENGINE: SHAP Explainable AI Drawer Evidence Shield

WHAT IT DOES: Answers the question WHY WAS THIS FLAGGED by popping open a side drawer that lists line-by-line evidence and legal rule citations.

HOW TO EXPLAIN: Think of an itemized restaurant or hospital bill. If a waiter charges you 5,000, you don't just pay blindly you want an itemized bill showing 2,000 for food, 1,500 for drinks, etc. PRASHNA-KAVACH

EXAMPLE: Pops open drawer showing exact GFR Rule 144 citation, CPWD cost variance, and matched duplicate project ID.

Module 4 SATARK-SIMULATION - - Pre-Approval What-If Real-Time Sandbox 15ms

CORE ENGINE: Sub-15ms Real-Time Sandbox

WHAT IT DOES: Allows officers to test a new project proposal title, cost, date, location in less than 15 milliseconds BEFORE signing sanction approvals.

HOW TO EXPLAIN: Like a Spell-Checker before sending an email or a Flight Simulator Before an MP or Collector signs off on spending 10 Lakhs of public money, they run a 3-millisecond simulation to check for anomalies

EXAMPLE: Officer inputs a proposed 4,95,000 community hall bill on March 29; Simulator flags March Rush + Tender-Split Evasion in 2ms.

Module 5 BHAVISHYA-REKHA - - Future Risk Trend Forecaster

CORE ENGINE: Linear Regression Time-Series Forecasting

WHAT IT DOES: Uses trend forecasting math to predict upcoming spending surges, March Rush spikes, and future quarterly risk trends.

HOW TO EXPLAIN: Like a Weather Forecast app It doesn't just tell you if it's raining right now it warns officers 3 months in advance that a heavy spending surge March Rush is coming so they can prepare early

EXAMPLE: Projects a 45% spike in unverified end-of-year fund sanctions for Q4, alerting auditors in January.

Module 6 SATARK-KARYA - - 1-Click Printable Legal Memo Builder

CORE ENGINE: Form GFR-19A Legal Memo Builder with SHA-256 Hash

WHAT IT DOES: Automatically converts AI findings into a ready-to-print official legal notice

HOW TO EXPLAIN: Like an automatic traffic challan / speeding ticket. Once the system

detects a violation, SATARK-KARYA generates an official legal document with 1 click that can be printed and handed to field officers

EXAMPLE: Generates confidential Memorandum MEMO/SATARK/2026/145555 with barcode and

District Magistrate countersignature block.

HIGH-LEVEL SYSTEM ARCHITECTURE LAYERS:

Layer 1: Real Government Data Warehouse: Ingests 100% real live data from official M

oSPI eSAKSHI portals 176,925 projects, 109,475 payment vouchers, 27,234 contractors across 37 States & UTs.

Layer 2: Dual-Brain Architecture Cold vs Hot Path: Heavy offline batch processing Cold

Path combined with sub-15ms live scoring gateway Hot Path.

Layer 3: 7 Independent AI Sentinels: Multi-layer defense inspecting statutory rules,

text duplicates, CPWD benchmarks, network cartels, 12D ML outliers, digit math, and satellite geotags.

Layer 4: Itemized Receipt Risk Scorer: 0 to 100 composite risk score with additive waterfall

breakdown replacing obscure black-box AI.

Layer 5: Role Dashboards 1-Click Legal Notice: Custom views for MP, District Collector,

State Nodal, and Ministry with Form GFR-19A printable memorandums.

30-SECOND QUICK ELEVATOR PITCH:

SATARKs core architecture has 6 main engines: DRISHTI gives a 0-100 report card score

, SAMVAAD lets you ask questions like ChatGPT, PRASHNA-KAVACH gives itemized proof of why a project was flagged, S